

CODE SPORT



In this month's column, we continue to discuss a medley of computer science interview questions.

We have been discussing computer science interview questions in the last few columns. We continue this discussion in this month's column, focusing on machine learning. One of our readers has requested that solutions to the previous month's questions be given in the present column. I would like to give some more time to our readers to solve the questions given last month. We will discuss the solutions to these problems in next month's column. So please do share your solutions to these questions. I would like to thank Pranav Bhargava who sent in the correct solutions to eight of the questions in last month's column. Let us get started on this month's questions.

1. What is the structured prediction problem in machine learning? Can you give an example of this? What methods will you use for solving structured prediction?
2. You have built a B2C (business to consumer) startup, wherein you offer health advice to your consumers based on their profile, behavioural data collected daily, etc. You have been able to sign up 10,000 customers to your platform. However, for the past two weeks, you have been finding a number of customers dropping off from your platform. In order to avoid further loss of customers, you want to predict which customers are unhappy on your platform and hence are likely to leave next. This is the churn prediction problem, where you want to predict customers likely to churn. Will you model this as a linear regression problem or a binary classification problem (likely to leave or not)? What kind of features will you use for the churn prediction problem?
3. Brand monitoring is an important business activity. Businesses monitor their social media channels to get an understanding of what their customers are thinking about the brand. They also monitor the social media channels of competitors. As a business owner, you will want to know when/where/how your brand is being mentioned, both explicitly (e.g., when someone mentions 'Google') and implicitly (e.g., when someone says 'the search giant'), as well as the sentiment associated with it. If there's a sudden surge of negative sentiment in your brand mentions, you will want to do something about it as soon as possible. Sentiment analysis is a typical ML task. Given the huge social media presence, automated tools are needed for it. Let us restrict the problem to one social media platform, namely, Twitter. You want to build a brand monitoring tool for Twitter posts, reporting the aggregated score daily (either positive or negative), and you also want to flag tweets that are highly positive and highly negative. Can you explain how you will build an automated tool using ML for this problem?
4. You find a tweet on an airline brand with the following text: "Thank you so much XXX Airlines!!! You sent my luggage to SFO when I am travelling to Chicago!!! Hope I will get it back by at least next month." You find that your solution flagged it as a positive tweet mistakenly. This tweet, while sounding positive on the surface, is an example of subtle sarcasm. How will you tweak your solution in (3) to handle such cases?
5. You are building a content sharing platform, and one of its features is to provide a news feed personalised for each user based on his profile. Let us simplify the problem. You have a list of news items given, and you need to rank the items for each user so that the top 100 items shown to him are personalised to his profile.